ModelIt!

Student Activity Pack

# Why Are There Seashells on Mountains?

The Mystery of Fossils Found in Surprising Places

Grade 4 | NGSS 4-ESS1-1 | Patterns

STUDENT NAME _____ DATE _____



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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever found a fossil or seen one in a museum? What did it look like?**

2. **How do you think fossils get inside of rocks?**

3. **If you found a seashell on top of a mountain, how would you explain it?**

4. **Draw what you think the layers of rock under the ground look like.**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Sediment Buildup | Earth Movement Speed | Fossil Exposure

| EXTERNAL | INTERNAL |
|---|---|
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Ocean Floor

High sediment, no Earth movement

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Mountain Building

Earth Movement Speed set to high

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Erosion Reveal

Moderate sediment and moderate movement

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Erosion Rate** — Wind and rain wear away mountain surfaces, revealing deeper and older fossil layers

**Volcanic Activity** — Volcanic ash creates new layers that can preserve fossils in fine detail

**Climate Changes** — Different climates create different types of sediment and preserve different organisms

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**How did seashells and ocean fossils end up on top of a mountain thousands of feet above the sea?** Use evidence from your simulation to support your answers.

**1 If a mountain has ocean fossils at the top, what was that land like millions of years ago?**

**2 Why are the oldest fossils usually found in the deepest rock layers?**

**3 How is finding a fossil like reading a page from Earth's history book?**

**4 What could happen to fossils if erosion wears away the rock they are trapped in?**

**5 How does your model help explain something that took millions of years to happen?**

# 8 STEM Challenge: Design a Fossil Dig Site

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## YOUR ENGINEERING MISSION

Design a model showing how fossils from different time periods end up in different rock layers, and plan a dig to find them.

### THE SCENARIO

A museum has hired your team to plan a fossil dig in the mountains. You need to predict which rock layers will contain which types of fossils, and design a dig plan that carefully removes layers without damaging the fossils below.

### Engineering Design Thinking

1. HOW MANY ROCK LAYERS WILL YOUR MODEL SHOW, AND WHAT FOSSILS ARE IN EACH?

2. WHAT TOOLS WOULD A REAL PALEONTOLOGIST USE TO CAREFULLY REMOVE ROCK FROM FOSSILS?

3. HOW WILL YOU LABEL EACH LAYER WITH ITS AGE AND THE TYPE OF ENVIRONMENT IT CAME FROM?

4. WHAT SAFETY RULES SHOULD YOUR DIG TEAM FOLLOW TO PROTECT THE FOSSILS?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Paleontologists study fossils to understand ancient life and how Earth has changed. They dig up dinosaur bones, ancient plants, and ocean creatures found in rock. They earn $55,000–$100,000/year.